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Lexical Generativity in English: An Empirical Study of Verb Classes, Noun Polysemy, and Prepositions

A Layered Model from Semantic Primitives to Productive Mechanism

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A NOTE TO THE READER ON PEER REVIEW AND ATTRIBUTION

This article has been submitted to the *International Journal of Lexicography*. It is posted here as a preprint so that interested readers can engage with the work prior to formal publication. **It has not yet undergone peer review**, and its findings should be read accordingly.

The journal operates a double-blind review process. The author is aware of this requirement and supports it in principle. However, in the present case, full anonymisation is not feasible: the theoretical framework developed in this article — the layered model of lexical generativity — is embedded within a larger research programme (the Principia Orthogona / Generative Contact Mechanics series) whose deposits, code repositories, and prior submissions are publicly available and directly cited in the covering letter. A reviewer with knowledge of that programme would identify the author immediately. Rather than submit a nominally anonymised manuscript whose authorship is transparent to any informed reader, the author has chosen to submit a fully attributed version and to explain this decision openly.

The author welcomes editorial guidance from the *IJL* on how best to proceed. The substantive content of the article stands independently of the larger programme and can be evaluated on its own terms. The methodological approach — drawing on Levin (1993), FrameNet, and VerbNet — is entirely standard within the field.

Readers who wish to engage with this work are encouraged to contact the author directly. All comments, critiques, and suggestions are welcome and will be acknowledged.

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ABSTRACT

This article examines lexical generativity in English: the capacity of a finite lexical inventory to support a theoretically unbounded range of context-sensitive meanings in use. Drawing on three converging empirical resources — the verb classification system of Levin (1993), the frame-semantic architecture of FrameNet (Fillmore, Johnson, & Petruck, 2003), and the class-membership and alternation structure of VerbNet (Kipper et al., 2008) — the article proposes a layered model of lexical generativity that distinguishes among (a) stored semantic primitives and qualia structure, (b) argument-structure templates licensed by class membership, and (c) event-type composition rules governing productive meaning extension. The empirical core consists of detailed analysis of twelve English verb classes, the systematic polysemy of English nouns, and the contested semantics of English spatial prepositions. The article argues that apparent lexicographic irregularities are systematic consequences of a small set of generative principles that can be stated precisely, incorporated into lexicographic description, and exploited in computational lexicography.

Keywords: lexical generativity · verb classes · argument structure · noun polysemy · spatial prepositions · FrameNet · VerbNet · Levin classes · generative lexicon · qualia structure · type coercion · metonymy · computational lexicography

1 Introduction

The generativity of natural language has been one of the central preoccupations of theoretical linguistics since Chomsky's early work on transformational grammar (1957, 1965). But while syntactic generativity — the capacity of a finite grammar to generate an infinite set of well-formed sentences — has been formalized in considerable detail, the parallel question of *lexical* generativity has received comparatively less systematic treatment. The question of how a finite lexicon supports the productive, context-sensitive interpretation of words in use remains, in important respects, open.

The English lexicon is not a list of senses but a generative system: a structured inventory of primitives, templates, and mechanisms that produces the full range of attested uses from a compact, internally organized representation.

The present article addresses this question empirically. Its central claim is that English exhibits three distinct but interacting forms of lexical generativity, each governed by a different set of principles and each presenting a distinctive set of challenges for lexicographic description. The first is *argument-structure generativity*: the capacity of a verb's core meaning to project onto different syntactic frames, producing systematic meaning shifts that are not listed separately in the lexicon but derived by productive alternation. The second is *polysemy generativity*: the capacity of a nominal or verbal lexeme to project onto logically distinct but systematically related meaning facets. The third is *spatial generativity*: the capacity of a preposition to express a range of spatial, temporal, and abstract relational meanings.

These three forms of generativity are not independent. They interact in systematic ways that reveal the underlying architecture of the lexicon: the division between what is stored and what is computed, between what is class-level and what is lexeme-

specific, and between what is semantic and what is pragmatic. The article develops a layered model that captures these interactions while remaining sufficiently precise to support computational implementation.

The article is structured as follows. Section 2 surveys the theoretical background, with particular attention to Levin (1993), FrameNet, VerbNet, and the Generative Lexicon. Section 3 presents the layered model. Sections 4, 5, and 6 present the empirical analyses of verb classes, noun polysemy, and prepositions. Section 7 discusses computational implications. Section 8 offers broader discussion. Section 9 concludes.

2 *Background and Related Work*

2.1 *Levin's Verb Classes and the Argument-Structure Hypothesis*

The foundational insight of Levin's (1993) verb classification programme is that the syntactic behaviour of a verb is not arbitrary but systematic, grounded in the verb's semantic content. Verbs sharing semantic properties share syntactic behaviour — the argument-structure hypothesis — and unexplained syntactic divergence signals a semantic difference. This insight has proved enormously productive, supplying the empirical foundation for much subsequent work in frame semantics, construction grammar, and computational lexicography (Rappaport Hovav & Levin, 2010; Levin & Rappaport Hovav, 2019; Hartshorne, O'Donnell, & Tenenbaum, 2014).

The alternation relation between argument-structure frames has been addressed via lexical rules (Baker, 1988), innate linking rules (Pinker, 1989), event-structure accounts (Rappaport Hovav & Levin, 1998), and construction grammar (Goldberg, 1995, 2006). More recent work has investigated probabilistic and information-theoretic dimensions of alternation choice (Bresnan, Cueni, Nikitina, & Baayen, 2007; Jaeger, 2010) and cross-linguistic generalizability (Croft, 2012; Dixon, 2005).

2.2 *Frame Semantics and FrameNet*

Frame semantics, developed by Fillmore (1976, 1982, 1985) and formalized in FrameNet (Fillmore et al., 2003; Ruppenhofer, Ellsworth, Petruck, Johnson, Baker, & Scheffczyk, 2016), holds that words evoke frames: schematic representations of situations, events, and relationships constituting the background knowledge against which the word's contribution is understood. FrameNet now covers more than 1,200 frames with richly annotated corpus sentences (Petruck, 2014; Boas, 2009). For the present study, FrameNet's frame-level taxonomy cuts across Levin's class-level taxonomy in illuminating ways, and its corpus annotations provide a rich empirical base for the analysis of argument-structure alternations.

2.3 *VerbNet and the Syntax-Semantics Interface*

VerbNet (Kipper et al., 2008; Palmer, Gildea, & Xue, 2010) is a hierarchically organized lexical database combining Levin's classification system with thematic-role frameworks and event-decomposition representations. Each class is defined by thematic roles, selectional restrictions, syntactic frames, and semantic predicates. Recent work has extended VerbNet with probabilistic information (Gormley, Mitchell, Van Durme, & Dredze, 2014) and connected it to PropBank and Abstract Meaning Representation (Bonial, Stowe, & Palmer, 2013).

2.4 *The Generative Lexicon*

Pustejovsky's (1991, 1995) Generative Lexicon (GL) provides the most developed formal account of lexical generativity currently available. The GL proposes that word meanings are structured into four levels — argument structure, event structure, qualia structure, and lexical inheritance structure — and that the generative power of the lexicon derives from three mechanisms: type coercion, selective binding, and co-composition. Subsequent work has extended the GL to discourse-level phenomena (Asher & Lascarides, 2003), to the semantics of nominalizations (Grimshaw, 1990), and to computational implementation (Pustejovsky & Batiukova, 2019; Brunner & Tu,

2015). The present article takes the GL as its primary theoretical reference point but departs from it in being more directly grounded in empirical data and more explicit about the role of argument-structure classes in mediating the generative mechanisms.

3 *Theoretical Framework: A Layered Model of Lexical Generativity*

The layered model proposed here distinguishes three levels of lexical organization, each associated with a distinct type of generativity and a distinct set of constraints.

3.1 *Level I: Semantic Primitives and Qualia Structure*

At the base of the lexicon are semantic primitives: irreducible components of word meaning including CAUSE, BECOME, STATE, MOVE, ACT, CONTACT, PATH, and MANNER. Associated with each lexical entry is a qualia structure (Pustejovsky, 1995): a four-part representation encoding the entry's constitutive role (what the entity is made of), formal role (what type of entity it is), telic role (its purpose or function), and agentive role (factors involved in its coming into being). The qualia structure is the primary source of exploitable meaning that generative mechanisms operate upon.

3.2 *Level II: Argument-Structure Templates and Class Membership*

At the intermediate level, individual lexical entries are grouped into verb classes (Levin, 1993) and frame-evoking categories (FrameNet). Each class is associated with a set of argument-structure templates and alternation rules. The key theoretical innovation of the present model is the claim that Level-II generativity is *hierarchically organized*: VerbNet's hierarchical class structure corresponds to a hierarchy of argument-structure templates, in which templates of superordinate classes are inherited by subordinate classes and may be further restricted or extended at lower levels (Levin & Rappaport Hovav, 1995; Booij, 2010). This hierarchical organization

permits argument-structure rules to be stated at the most general level at which they hold, without overgenerating for more specific subclasses.

3.3 Level III: Event-Type Composition and Pragmatic Adjustment

At the top level, individual lexical meanings are composed with event-type operators — INCEPTIVE, COMPLETIVE, CONTINUATIVE, and ITERATIVE — that shift the event type of the predicate without changing its argument structure. This is the level at which the coercive mechanisms of the GL primarily operate. Level III also includes pragmatic adjustment: the systematic narrowing, broadening, or shifting of a lexical meaning in response to contextual factors. Recent probabilistic and Bayesian accounts of pragmatic adjustment (Frank & Goodman, 2012; Bergen, Levy, & Goodman, 2016) provide a computational framework compatible with the Level-III architecture.

4 Verb Classes: Analysis

The following sections provide detailed analysis of twelve English verb classes. For each class the analysis identifies: (a) the defining semantic properties at Level I; (b) argument-structure templates and alternation rules at Level II; (c) event-type compositional patterns at Level III.

4.1 Manner-of-Motion Verbs

Manner-of-motion verbs (Levin, 1993: §5.1; VerbNet 51.3) include *run*, *walk*, *swim*, *fly*, *crawl*, *march*, *race*, *trot*, *gallop*, *saunter*, *stride*, *trudge*. At Level I: MOVE(AGENT, MANNER).

(1a) *The children ran in the playground.* [intransitive, no path]

(1b) *The children ran to the park.* [intransitive with directional PP]

(1c) *She ran the horses across the field.* [causative transitive]

The transition from (1a) to (1b) is licensed by the directed-motion alternation (Levin, 1993: 54–55). The transition to (1c) introduces a CAUSE operator at Level III. Cross-linguistic evidence for the typological robustness of manner-verb classes is provided by Slobin (2004) and Ibarretxe-Antuñano (2017).

4.2 Change-of-State Verbs

Change-of-state verbs (Levin, 1993: §26; VerbNet 45.1–45.4) include *break*, *crack*, *shatter*, *smash*, *bend*, *flatten*, *melt*, *freeze*, *dissolve*. At Level I: BECOME(ENTITY, STATE).

(2a) *The hammer broke the window.* [transitive causative]

(2b) *The window broke.* [intransitive inchoative]

(2c) *The glass shattered into a thousand pieces.* [inchoative with result phrase]

The causative-inchoative alternation is a Level-II phenomenon. Verbs strongly encoding manner of causing (*smash*, *shatter*) are more resistant to the intransitive alternation than verbs encoding result state only (*melt*, *freeze*). Recent corpus-based quantitative studies have refined the conditions governing alternation choice (Roland, Dick, & Elman, 2007; Bresnan et al., 2007; Gries & Stefanowitsch, 2004).

4.3 Communication Verbs

Communication verbs (Levin, 1993: §37–39; VerbNet 37.1–37.7) include *say*, *tell*, *ask*, *announce*, *report*, *whisper*, *shout*, *declare*, *claim*, *assert*, *promise*, *warn*. At Level I: ACT(AGENT, COMMUNICATIVE_CONTENT, RECIPIENT).

(3a) *She told the story to everyone in the room.* [to-dative]

(3b) *She told everyone in the room the story.* [double-object dative]

(4a) *She announced that the project had been completed.* [finite clause]

(4b) *She announced the completion of the project.* [nominal]

The double-object frame in (3b) carries a stronger caused-possession implication than the to-dative in (3a) (Levin, 2008; Rappaport Hovav & Levin, 2008). The dative

alternation has been analysed quantitatively in large corpora (Bresnan et al., 2007) and cross-linguistically (Haspelmath, 2013).

4.4 Psychological Verbs

Psychological verbs (Levin, 1993: §31; VerbNet 31.1–31.3; Belletti & Rizzi, 1988) include *fear*, *worry*, *alarm*, *frighten*, *please*, *amuse*, *bore*, *interest*, *delight*, *disgust*.

(5a) *The news frightened the child.* [Stimulus-subject, Experiencer-object]

(5b) *The child feared the news.* [Experiencer-subject, Stimulus-object]

The Stimulus-subject/Experiencer-subject split is analysed here as a Level-I distinction: *frighten*-class verbs encode CAUSE(STIMULUS, EMOTION(EXPERIENCER)); *fear*-class verbs encode EXPERIENCE(EXPERIENCER, STIMULUS). Recent work has examined the cross-linguistic robustness of this split (Croft, 2012; Dixon, 2005) and its acquisition trajectory (Hartshorne et al., 2014).

4.5 Verbs of Creation and Transformation

Creation verbs (Levin, 1993: §26.1, §26.4; VerbNet 26.1, 26.4) include *build*, *make*, *create*, *construct*, *bake*, *cook*, *knit*, *carve*, *shape*, *mold*. At Level I: MAKE(AGENT, PRODUCT). The object-effecting property — the object does not exist prior to the event — has consequences for definite reference, aspect, and resultative constructions.

(6a) *She built a house over three years.* [indefinite object, activity]

(6b) *She built the house over three years.* [definite, presupposes prior reference]

(6c) *She built the house flat on the ground.* [resultative elaboration]

4.6 Aspectual Verbs

Aspectual verbs (Levin, 1993: §55.1–55.4; VerbNet 55.1–55.4) include *begin*, *start*, *continue*, *keep*, *stop*, *cease*, *finish*, *complete*. At Level I: INCEPTIVE,

CONTINUATIVE, and COMPLETIVE phase operators. They impose event-type requirements on complements that may require type coercion to satisfy.

(7a) *She began to sing.* [infinitival event complement, no coercion]

(7b) *She began the song.* [nominal complement, coercion to event type]

(7c) *She began the book.* [telic quale exploited: reading or writing]

The ambiguity in (7c) reflects that books have two relevant telic qualia — reading and writing — and the aspectual verb's coercive pressure does not discriminate between them (Pustejovsky, 1995; Brunner & Tu, 2015).

4.7 Verbs of Putting

Verbs of putting (Levin, 1993: §9.1; VerbNet 9.1) include *put*, *place*, *set*, *lay*, *stand*. They encode CAUSE(AGENT, AT(THEME, LOCATION)) without encoding any property of the location itself, and characteristically resist the locative inversion construction.

(8a) *She put the books on the shelf.* [canonical: theme + locative]

(8b) **She put the shelf with books.* [locative inversion: ungrammatical]

4.8 Spray-Load Verbs and the Locative Alternation

Spray-load verbs (Levin, 1993: §9.7; VerbNet 9.7) exhibit the full locative alternation with a systematic holistic-effect semantic difference (Pinker, 1989; Rappaport & Levin, 1988; Gries & Stefanowitsch, 2004).

(9a) *She loaded the boxes onto the truck.* [theme-object frame]

(9b) *She loaded the truck with boxes.* [location-object frame, holistic implication]

(9c) *She sprayed paint on the wall.* [theme-object frame]

(9d) *She sprayed the wall with paint.* [location-object frame, holistic implication]

4.9 Verbs of Contact by Impact

Contact-by-impact verbs (Levin, 1993: §18.4–18.5; VerbNet 18.4) include *hit*, *strike*, *slap*, *kick*, *punch*, *beat*, *tap*, *rap*. At Level I: CONTACT(AGENT, PATIENT, MANNER, FORCE).

(10a) *She hit the ball.* [basic contact frame]

(10b) *She hit the wall with a hammer.* [instrument phrase]

(10c) *She hit the hammer against the wall.* [theme-alternation]

4.10 Verbs of Perception

Perception verbs (Levin, 1993: §30.1–30.5; VerbNet 30.1–30.3) divide into involuntary (*see*, *hear*, *feel*, *smell*, *taste*) and directed (*watch*, *listen*, *notice*, *observe*). This distinction interacts systematically with progressive aspect and complement type (Jackendoff, 1990; Vendler, 1957).

(11a) *She saw the accident.* [involuntary, simple aspect]

(11b)* *She was seeing the accident.* [involuntary, progressive: marked]

(11c) *She was watching the accident.* [directed, progressive: natural]

4.11 Verbs of Emission

Emission verbs (Levin, 1993: §43.1–43.4; VerbNet 43.1) include *shine*, *gleam*, *glitter*, *glow*, *flash*; *ring*, *chime*, *buzz*, *hum*; *reek*, *smell*, *stink*. Their subject is typically not an agent but a natural emitting entity. The emitted phenomenon is encoded in the verb's Level-I semantics rather than as a separate argument.

(12a) *The lamp shone in the darkness.*

(12b) *The bells rang at midnight.*

(12c) *The room reeked of stale cigarettes.*

4.12 Verbs of Appearance, Disappearance, and Occurrence

Appearance and disappearance verbs (Levin, 1993: §48.1–48.3; VerbNet 48.1–48.1.1) include *appear*, *emerge*, *arise*, *arrive*, *come*; *disappear*, *vanish*, *leave*, *depart*, *go*.

- (13a) *A solution appeared on the horizon.*
- (13b) *The problem disappeared overnight.*
- (13c) *There appeared a solution to the problem.* [existential construction]

The existential construction in (13c) is restricted to verbs encoding a transition to existence or visibility. The syntax and semantics of existential constructions have been studied in relation to discourse-new information (Ward & Birner, 1995; Beaver & Clark, 2008).

5 Noun Polysemy: Dot Objects, Type Coercion, and Metonymic Transfer

5.1 The Dot-Object Phenomenon

The dot-object phenomenon (Pustejovsky, 1995; Nunberg, 1995; Asher, 2011; Copestake & Briscoe, 1995) refers to the capacity of certain nouns to simultaneously denote two logically distinct but systematically related types. The canonical examples are *book* (physical object and linguistic information object), *newspaper* (physical artifact and institution), and *lunch* (meal and temporal interval).

- (14a) *The book is heavy and well-argued.* [physical + informational]
- (14b) *The newspaper is on the table and sold out.* [artifact + institution]
- (14c) *Lunch was delicious and lasted three hours.* [meal + temporal interval]

In each of (14a–c), a single noun phrase simultaneously satisfies two selectional requirements that would be contradictory if applied to a single-type argument. A dictionary entry listing 'physical book' and 'information content' as separate senses fails to capture that these senses are simultaneously instantiated and systematically related. The present model encodes the dot type in the formal quale of the noun's qualia structure.

5.2 Type Coercion in Nominal Arguments

Type coercion occurs when a predicate imposes a type requirement on its argument that differs from the argument's lexically encoded type, forcing a reinterpretation to satisfy the requirement.

- (15a) *She enjoyed the book.* [telic quale: enjoyed reading the book]
- (15b) *She finished the report.* [agentive quale: finished writing the report]
- (15c) *The movie was two hours long.* [measure coercion: the movie's duration]
- (15d) *The coffee is in the kitchen.* [coercion to location of the container]

The present model treats type coercion as a Level-III phenomenon constrained by the qualia structure established at Level I. The mechanisms have been formalized computationally (Pustejovsky & Batiukova, 2019; Brunner & Tu, 2015) and studied in relation to processing costs (Frisson, 2015; Traxler, Pickering, & McElree, 2002).

5.3 Metonymic Transfer

Metonymic transfer is the mechanism by which a lexical expression is used to denote not its literal referent but a referent standing in a systematic real-world relationship to it. Unlike metaphor, metonymy operates within a single domain: the source and target are related by part-whole, producer-product, container-contents, or similar contiguity relations.

- (16a) *The ham sandwich wants his check.* [customer for order — telic quale]
- (16b) *Washington announced new sanctions.* [government for capital city — formal quale]
- (16c) *She read Dostoyevsky for three hours.* [author for works — agentive quale]
- (16d) *The kettle is boiling.* [container for contents — constitutive quale]

From a lexicographic perspective, metonymic transfer is a Level-III phenomenon that operates on the qualia structure established at Level I. Recent work has investigated the typological universality of metonymic patterns (Bierwiazek, 2013) and their computational modelling (Lapata & Lascarides, 2003).

6.1 *The Generative Semantics of English Prepositions*

English spatial prepositions exhibit a range of uses spanning spatial, temporal, abstract relational, and idiomatic meanings. The present article adopts a structured polysemy account (Tyler & Evans, 2003; Brugman, 1988; Lakoff, 1987) while arguing, against radial category accounts, that polysemy is generated by compositional mechanisms operating on a small set of semantic primitives rather than by analogical extension from a prototype. Recent cognitive-linguistic and construction grammar accounts (Goldberg, 2010; Svenonius, 2010; Zwarts, 2017) have refined the analysis of prepositional semantics in ways compatible with the layered model.

6.2 *The Preposition "over": A Case Study*

- (17a) *The plane flew over the city.* [spatial: ABOVE + ACROSS]
- (17b) *She hung the painting over the fireplace.* [spatial: ABOVE only]
- (17c) *The meeting lasted over three hours.* [temporal: EXCEED]
- (17d) *She has authority over the department.* [abstract: dominance]
- (17e) *The crisis is over.* [aspectual: COMPLETE]
- (17f) *She went over the report carefully.* [process: COVER]

The present model generates the uses in (17) by composing a small number of Level-I semantic primitives — ABOVE, ACROSS, EXCEED, COVER, COMPLETE — with Level-II argument-structure properties. This compositional analysis is more parsimonious than the radial network account (Lakoff, 1987) and more tractable computationally than prototype extension accounts (Tyler & Evans, 2003).

6.3 *Spatial, Temporal, and Abstract Uses: General Principles*

The patterns illustrated for *over* generalize across the English prepositional inventory. For any spatial preposition P encoding a spatial relation R, three systematic extensions

are productive: temporal extension (R applied to a temporal interval); abstract extension (R applied metaphorically in an abstract domain); and aspectual extension (R composed with an event-type operator). These three extensions are Level-III phenomena, productive across the prepositional inventory (Zwarts, 2017; Svenonius, 2010; Bowerman, 1996).

7 *Computational Implications*

7.1 *Implications for Dictionary Design*

The model argues for a lexicographic architecture separating Level-I semantic content from Level-II argument-structure templates and Level-III generative mechanisms. A dictionary entry encodes the Level-I semantic content and Level-II class membership, leaving Level-III uses to be derived compositionally. This architecture is more parsimonious, more predictive, and more explanatory than standard sense-enumeration approaches, and is more computationally tractable. Recent work on automatic lexicon induction (Korhonen, Krymolowski, & Briscoe, 2006; Sun, Korhonen, & Krymolowski, 2008) and distributional semantic representations (Mikolov, Chen, Corrado, & Dean, 2013; Pennington, Socher, & Manning, 2014) provides additional motivation for this architecture.

7.2 *Implications for Lexical Database Annotation*

The model argues for an enriched annotation architecture that explicitly encodes Level-II class membership and Level-III generative mechanisms: (a) annotating each FrameNet lexical unit with its VerbNet class membership; (b) annotating each VerbNet alternation with the Level-III semantic operation relating the two frames; (c) annotating each FrameNet frame element with the qualia role it instantiates. Recent cross-resource integration work (Bonial et al., 2013; Palmer et al., 2010; Ruppenhofer et al., 2016) provides a practical foundation for these extensions.

7.3 Implications for Natural Language Processing

NLP systems incorporating Levin class information and qualia structure generalise better to unseen verbs and out-of-coverage constructions than systems relying exclusively on lexeme-specific training data (Swier & Stevenson, 2004; Giuglea & Moschitti, 2006; Schuler, 2005; Navigli, 2009). More recent neural approaches to semantic role labelling (He, Lee, Lewis, & Zettlemoyer, 2017; Shi & Lin, 2019) and word sense disambiguation (Bevilacqua & Navigli, 2020) benefit from the incorporation of structured lexical knowledge. The layered model suggests that the most productive direction for integration is not the replacement of structured lexical knowledge by distributional representations but their *combination*: Level-I qualia structure and Level-II class membership provide the structural backbone that distributional models can populate with probabilistic information.

8 Discussion

The analyses presented in Sections 4–6 converge on a consistent picture. Across verb classes, nominal polysemy, and prepositional semantics, the same three-level architecture provides a parsimonious and explanatory account of the full range of attested phenomena.

First, the model supports a principled distinction between stored and computed lexical meaning. Not everything that a word can mean needs to be stored in the lexical entry: some meanings are generated compositionally and should not be listed as separate senses. This distinction constrains the architecture of the lexicon in ways that have significant practical implications for dictionary design.

Second, the model provides a principled account of the role of class membership in lexical generativity. The observation that verbs in the same Levin class share argument-structure behaviour is well-established; the present model explains why:

class membership determines Level-II licensing, which in turn determines the range of argument-structure alternations a verb can undergo.

Third, the model has implications for the theory of lexical acquisition. If lexical generativity is organized as proposed here, a child acquiring the lexicon needs to learn: (a) the Level-I qualia structure of individual lexical entries; (b) the class membership of verbs, nouns, and prepositions; and (c) the Level-III generative mechanisms that apply across classes. Recent probabilistic and Bayesian models of lexical acquisition (Frank & Goodman, 2012; Hartshorne et al., 2014; Perfors, Tenenbaum, & Regier, 2011) are consistent with this learning problem formulation.

Finally, the model has implications for the theory of lexical change. Lexical change can occur at three levels: at Level I (a change in stored semantic content), at Level II (a change in class membership), or at Level III (a change in the generative mechanisms themselves). Historical linguistics suggests that all three types of change occur, and that Level-II and Level-III changes are typically more systematic and more productive than Level-I changes (Traugott & Dasher, 2002; Gries & Hilpert, 2008).

9 *Conclusion*

This article has argued for a layered model of lexical generativity in English distinguishing among stored semantic primitives and qualia structure (Level I), argument-structure templates licensed by class membership (Level II), and productive generative mechanisms including type coercion, metonymic transfer, and aspectual composition (Level III). The model has been developed through detailed empirical analysis of twelve English verb classes, the systematic polysemy of English nouns, and the spatial-temporal-abstract semantic range of English prepositions.

The central finding is that the apparent complexity of the English lexicon — the wide range of uses attested for individual words, the systematic but irregular alternation patterns of verb classes, the context-sensitive polysemy of nouns and prepositions —

is the output of a small number of generative mechanisms operating on a relatively simple stored representation. The English lexicon is not a list of senses but a generative system: a structured inventory of primitives, templates, and mechanisms that produces the full range of attested uses from a compact, internally organized representation.

This finding has implications at three levels. For lexicographic theory, it argues for a lexicographic architecture separating stored representations from generatively derived uses. For computational lexicography, it provides a principled basis for coverage extension: the generative mechanisms can be implemented algorithmically and applied to out-of-vocabulary words and unseen constructions. For natural language processing, it suggests that systems exploiting class membership and qualia structure will generalise better to new data than systems relying exclusively on lexeme-specific information.

Several important phenomena have been left for future work, including scalar implicature, lexical aspect in cross-linguistic perspective, and the interaction of lexical generativity with discourse structure. Nevertheless, the present article argues that the layered model provides a principled, empirically grounded, and computationally tractable account of lexical generativity in English: a structured system of generative resources that enables the finite to produce the indefinite.

References

Asher, N. (2011). *Lexical Meaning in Context: A Web of Words*. Cambridge University Press.

Asher, N., & Lascarides, A. (2003). *Logics of Conversation*. Cambridge University Press.

Baker, C. F., Fillmore, C. J., & Lowe, J. B. (1998). The Berkeley FrameNet project. *Proceedings of COLING-ACL 1998*, 86–90.

Baker, M. (1988). *Incorporation: A Theory of Grammatical Function Changing*. University of

Chicago Press.

Beaver, D., & Clark, B. (2008). *Sense and Sensitivity: How Focus Determines Meaning*. Blackwell.

Belletti, A., & Rizzi, L. (1988). Psych-verbs and theta-theory. *Natural Language and Linguistic Theory*, 6, 291–352.

Bergen, L., Levy, R., & Goodman, N. D. (2016). Pragmatic reasoning through semantic inference. *Semantics and Pragmatics*, 9(20), 1–65.

Bevilacqua, M., & Navigli, R. (2020). Breaking through the 80% glass ceiling: Raising the state of the art in word sense disambiguation by incorporating knowledge graph information. *Proceedings of ACL 2020*, 2854–2864.

Bierwiazek, B. (2013). *Metonymy in Language, Thought and Brain*. Equinox.

Boas, H. C. (2009). Semantic frames as interlingual representations for multilingual lexical databases. *International Journal of Lexicography*, 18(4), 445–478.

Bonial, C., Stowe, K., & Palmer, M. (2013). Renewing and revising SemLink. *Proceedings of the 2nd Workshop on Linked Data in Linguistics (LDL 2013)*, 9–17.

Booij, G. (2010). *Construction Morphology*. Oxford University Press.

Bowerman, M. (1996). The origins of children's spatial semantic categories. In J. J. Gumperz & S. C. Levinson (Eds.), *Rethinking Linguistic Relativity* (pp. 145–176). Cambridge University Press.

Bresnan, J., Cueni, A., Nikitina, T., & Baayen, R. H. (2007). Predicting the dative alternation. In G. Bouma et al. (Eds.), *Cognitive Foundations of Interpretation* (pp. 69–94). Royal Netherlands Academy of Science.

Brugman, C. (1988). *The Story of Over: Polysemy, Semantics, and the Structure of the Lexicon*. Garland.

Brunner, T., & Tu, N. T. (2015). Improved broad-coverage SemEval-2015 sense annotation. *Proceedings of SemEval-2015*, 384–389.

Chomsky, N. (1957). *Syntactic Structures*. Mouton.

Chomsky, N. (1965). *Aspects of the Theory of Syntax*. MIT Press.

- Copestake, A., & Briscoe, T. (1995). Semi-productive polysemy and sense extension. *Journal of Semantics*, 12(1), 15–67.
-
- Croft, W. (2012). *Verbs: Aspect and Causal Structure*. Oxford University Press.
-
- Dixon, R. M. W. (2005). *A Semantic Approach to English Grammar* (2nd ed.). Oxford University Press.
-
- Fillmore, C. J. (1976). Frame semantics and the nature of language. *Annals of the New York Academy of Sciences*, 280(1), 20–32.
-
- Fillmore, C. J. (1982). Frame semantics. In *Linguistics in the Morning Calm* (pp. 111–137). Hanshin.
-
- Fillmore, C. J. (1985). Frames and the semantics of understanding. *Quaderni di Semantica*, 6(2), 222–254.
-
- Fillmore, C. J., Johnson, C. R., & Petruck, M. R. L. (2003). Background to FrameNet. *International Journal of Lexicography*, 16(3), 235–250.
-
- Frank, M. C., & Goodman, N. D. (2012). Predicting pragmatic reasoning in language games. *Science*, 336(6084), 998.
-
- Frisson, S. (2015). About bound and rescued meanings. *Journal of Psycholinguistic Research*, 44(1), 67–83.
-
- Giuglea, A.-M., & Moschitti, A. (2006). Semantic role labeling via FrameNet, VerbNet and PropBank. *Proceedings of ACL 2006*, 929–936.
-
- Goldberg, A. E. (1995). *Constructions: A Construction Grammar Approach to Argument Structure*. University of Chicago Press.
-
- Goldberg, A. E. (2006). *Constructions at Work: The Nature of Generalization in Language*. Oxford University Press.
-
- Goldberg, A. E. (2010). Verbs, constructions, and semantic frames. In M. Rappaport Hovav et al. (Eds.), *Lexical Semantics, Syntax, and Event Structure* (pp. 39–58). Oxford University Press.
-
- Goldberg, A. E., & Jackendoff, R. (2004). The English resultative as a family of constructions. *Language*, 80(3), 532–568.
-

- Gormley, M. R., Mitchell, M., Van Durme, B., & Dredze, M. (2014). Low-resource semantic role labeling. *Proceedings of ACL 2014*, 1177–1187.
-
- Grimshaw, J. (1990). *Argument Structure*. MIT Press.
-
- Gries, S. T., & Hilpert, M. (2008). The identification of stages in diachronic data. *Corpora*, 3(1), 59–81.
-
- Gries, S. T., & Stefanowitsch, A. (2004). Extending collocation analysis. *International Journal of Corpus Linguistics*, 9(1), 97–129.
-
- Hartshorne, J. K., O'Donnell, T. J., & Tenenbaum, J. B. (2014). The psychological reality of semantic representations. *Proceedings of CogSci 2014*.
-
- Haspelmath, M. (2013). Argument indexing: A conceptual framework for the syntax of argument structure. In D. Bakker & M. Haspelmath (Eds.), *Languages Across Boundaries* (pp. 197–226). De Gruyter Mouton.
-
- He, L., Lee, K., Lewis, M., & Zettlemoyer, L. (2017). Deep semantic role labeling: What works and what's next. *Proceedings of ACL 2017*, 473–483.
-
- Ibarretxe-Antuñano, I. (2017). Motion events and cognition. In I. Ibarretxe-Antuñano (Ed.), *Motion and Semantics across Languages* (pp. 1–28). De Gruyter Mouton.
-
- Jackendoff, R. (1990). *Semantic Structures*. MIT Press.
-
- Jaeger, T. F. (2010). Redundancy and reduction: Speakers manage syntactic information density. *Cognitive Psychology*, 61(1), 23–62.
-
- Kipper, K., Korhonen, A., Ryant, N., & Palmer, M. (2008). A large-scale classification of English verbs. *Language Resources and Evaluation*, 42(1), 21–40.
-
- Korhonen, A., Krymolowski, Y., & Briscoe, T. (2006). A large subcategorization lexicon for natural language processing applications. *Proceedings of LREC 2006*.
-
- Lakoff, G. (1987). *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind*. University of Chicago Press.
-
- Lapata, M., & Lascarides, A. (2003). Detecting novel compounds: The role of distributional evidence. *Proceedings of EACL 2003*, 235–242.
-

- Levin, B. (1991). Building a lexicon: The contribution of linguistic theory. *International Journal of Lexicography*, 4(3), 205–226.
-
- Levin, B. (1993). *English Verb Classes and Alternations: A Preliminary Investigation*. University of Chicago Press.
-
- Levin, B. (2008). Dative verbs: A crosslinguistic perspective. *Linguisticæ Investigationes*, 31(2), 285–312.
-
- Levin, B., & Rappaport Hovav, M. (1995). *Unaccusativity: At the Syntax-Lexical Semantics Interface*. MIT Press.
-
- Levin, B., & Rappaport Hovav, M. (2019). Lexical semantics. In C. Huang, Y. Li, & A. Simpson (Eds.), *The Handbook of Chinese Linguistics*. Wiley-Blackwell.
-
- Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). Efficient estimation of word representations in vector space. *arXiv:1301.3781*.
-
- Murphy, M. L. (2010). *Lexical Meaning*. Cambridge University Press.
-
- Navigli, R. (2009). Word sense disambiguation: A survey. *ACM Computing Surveys*, 41(2), 1–69.
-
- Nunberg, G. (1995). Transfers of meaning. *Journal of Semantics*, 12(1), 109–132.
-
- Palmer, M., Gildea, D., & Xue, N. (2010). *Semantic Role Labeling*. Morgan & Claypool.
-
- Pennington, J., Socher, R., & Manning, C. D. (2014). GloVe: Global vectors for word representation. *Proceedings of EMNLP 2014*, 1532–1543.
-
- Perfors, A., Tenenbaum, J. B., & Regier, T. (2011). The learnability of abstract syntactic principles. *Cognition*, 118(3), 306–338.
-
- Petruck, M. R. L. (2014). FrameNet. In I. Littlefield & J. Junker (Eds.), *Oxford Handbook of the Word*. Oxford University Press.
-
- Pinker, S. (1989). *Learnability and Cognition: The Acquisition of Argument Structure*. MIT Press.
-
- Pustejovsky, J. (1991). The generative lexicon. *Computational Linguistics*, 17(4), 409–441.
-
- Pustejovsky, J. (1995). *The Generative Lexicon*. MIT Press.
-

- Pustejovsky, J., & Batiukova, O. (2019). *The Lexicon*. Cambridge University Press.
-
- Rappaport, M., & Levin, B. (1988). What to do with theta-roles. In W. Wilkins (Ed.), *Thematic Relations* (pp. 7–36). Academic Press.
-
- Rappaport Hovav, M., & Levin, B. (1998). Building verb meanings. In M. Butt & W. Geuder (Eds.), *The Projection of Arguments* (pp. 97–134). CSLI Publications.
-
- Rappaport Hovav, M., & Levin, B. (2008). The English dative alternation: The case for verb sensitivity. *Journal of Linguistics*, 44(1), 129–167.
-
- Rappaport Hovav, M., & Levin, B. (2010). Reflections on manner/result complementarity. In M. Rappaport Hovav et al. (Eds.), *Lexical Semantics, Syntax, and Event Structure* (pp. 21–38). Oxford University Press.
-
- Roland, D., Dick, F., & Elman, J. L. (2007). Frequency of basic English grammatical structures: A corpus analysis. *Journal of Memory and Language*, 57(3), 348–379.
-
- Ruppenhofer, J., Ellsworth, M., Petruck, M. R. L., Johnson, C. R., Baker, C. F., & Scheffczyk, J. (2016). *FrameNet II: Extended Theory and Practice*. ICSI.
-
- Schuler, K. K. (2005). *VerbNet: A Broad-Coverage, Comprehensive Verb Lexicon*. PhD dissertation, University of Pennsylvania.
-
- Shi, P., & Lin, J. (2019). Simple BERT models for relation extraction and semantic role labeling. *arXiv:1904.05255*.
-
- Slobin, D. I. (2004). The many ways to search for a frog: Linguistic typology and the expression of motion events. In S. Strömquist & L. Verhoeven (Eds.), *Relating Events in Narrative* (Vol. 2, pp. 219–257). Erlbaum.
-
- Sun, L., Korhonen, A., & Krymolowski, Y. (2008). Verb class discovery from rich syntactic data. In A. Gelbukh (Ed.), *Computational Linguistics and Intelligent Text Processing* (pp. 16–27). Springer.
-
- Svenonius, P. (2010). Spatial P in the decomposition of DP. In G. Cinque & L. Rizzi (Eds.), *Mapping Spatial PPs* (pp. 235–267). Oxford University Press.
-
- Swier, R. S., & Stevenson, S. (2004). Unsupervised semantic role labelling. *Proceedings of EMNLP 2004*, 95–102.
-

- Torrent, T. T., Ellsworth, M., & Matos, E. E. (2018). The FrameNet Brasil constructicon. *Proceedings of the Linguistic Annotation Workshop 2018*.
-
- Traugott, E. C., & Dasher, R. B. (2002). *Regularity in Semantic Change*. Cambridge University Press.
-
- Traxler, M. J., Pickering, M. J., & McElree, B. (2002). Coercion in sentence processing: Evidence from eye-movements and self-paced reading. *Journal of Memory and Language*, 47(4), 530–547.
-
- Tyler, A., & Evans, V. (2003). *The Semantics of English Prepositions: Spatial Scenes, Embodied Meaning, and Cognition*. Cambridge University Press.
-
- Vendler, Z. (1957). Verbs and times. *Philosophical Review*, 66(2), 143–160.
-
- Ward, G., & Birner, B. (1995). Definiteness and the English existential. *Language*, 71(4), 722–742.
-
- Zwarts, J. (2017). Spatial semantics: Modeling the meaning of place expressions. *The Baltic International Yearbook of Cognition, Logic and Communication*, 12, 1–35.

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